

Cross-Domain Generalization and the Cardinality Curse:

A Multi-Benchmark Empirical Investigation of Prompt Tuning, Plain LoRA, Class-Specific Context, and Residual Tokens Across Agricultural Grading and Fine-Grained Object Recognition

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September 2026

Repository: <https://github.com/myavarih/CLIP-LoRA>

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1. Summary of Evaluated Runs

TABLE 1. Architectural and algorithmic specification of the four evaluated fine-tuning paradigms.

Run	Paradigm Name	Vision Adapter	Text Adapter	Prompt & Class Formulation	Loss Objective
Run 1	Infix CoOp + Ordinal	LoRA ($r=2, \alpha=1.0$)	Frozen Backbone	Infix context ($M=4$) + Learnable class tokens Δe_k	$\mathcal{L}_{ce} + \lambda_{ord}\mathcal{L}_{ord}$
Run 2	Plain Dual LoRA	LoRA ($r=2, \alpha=1.0$)	LoRA ($r=2, \alpha=1.0$)	Static natural template, frozen BPE token embeddings	Standard $\mathcal{L}_{ce}$
Run 3	CoOp-CSC Dual LoRA	LoRA ($r=2, \alpha=1.0$)	LoRA ($r=2, \alpha=1.0$)	Class-Specific Context ($K \times M$ continuous prompt vectors)	Standard $\mathcal{L}_{ce}$
Run 4	Plain LoRA + ResCls	LoRA ($r=2, \alpha=1.0$)	LoRA ($r=2, \alpha=1.0$)	Static natural template + Learnable residual class offset δ_k	Standard $\mathcal{L}_{ce}$

Note: I forgot to set the flag for Run 3 to use ordinal loss despite the plan.

TABLE 2. Benchmark datasets and experimental evaluation suite matrix (216 initial runs on seeds 11, 12, 13, plus 144 runs on seeds 21, 22, 23 with fixed data augmentation; 360 evaluated runs total).

Dataset	Classes (K)	Zero-Shot	Shots (k)	Seeds
Walnut (Initial Aug.)	6	22.26%	{1, 2, 4, 8, 16, 32}	11, 12, 13
Walnut (Fixed Aug.) [*]	6	22.26%	{1, 2, 4, 8, 16, 32}	21, 22, 23
Piarom (Initial Aug.)	5	24.91%	{1, 2, 4, 8, 16, 32}	11, 12, 13
Piarom (Fixed Aug.) [*]	5	24.91%	{1, 2, 4, 8, 16, 32}	21, 22, 23
Cars	196	65.54%	{1, 2, 4, 8, 16, 32}	11, 12, 13

^{*}*Note: Data augmentation was changed (fixed) in the 21, 22, 23 seed runs (letterbox square padding to eliminate object cropping, vertical flipping, subtle translation affine shift, and impulse/Gaussian noise).*

2. Multi-Dataset Benchmark Results

2.1. Cross-Dataset Performance Trajectory

Figure 1 illustrates the test accuracy progression across shot counts ($k \in \{1, 2, 4, 8, 16, 32\}$) across benchmarks, including the newly evaluated Piarom Date Shape and Walnut Quality Grading suites with fixed non-cropping augmentations.

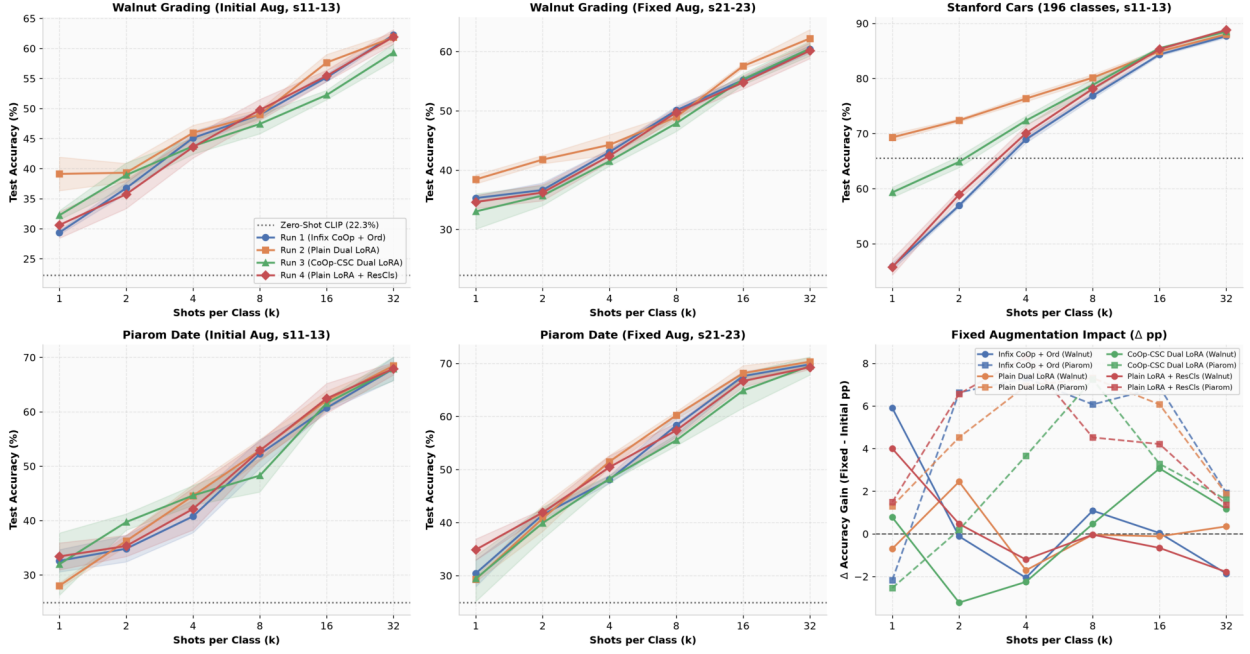


FIGURE 1. Few-shot test accuracy trajectories across 1 to 32 shots on Walnut under initial and fixed data augmentations ($K = 6$), Piarom Date Shape under initial and fixed data augmentations ($K = 5$), and Stanford Cars ($K = 196$). Shaded bands represent ± 1 standard deviation. The dashed line marks the frozen Zero-Shot CLIP baseline.

2.2. The Numbers!

Tables 3, 4, 5, 6, and 7 record the exact numerical test accuracy (mean $\pm$ standard deviation across seeds) for each paradigm and shot level. The top-performing method at each shot is highlighted in green.

TABLE 3. **Walnut Quality Grading — Initial Augmentation** ($K = 6$ classes, Seeds 11, 12, 13). Zero-Shot CLIP Baseline: **22.26%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	29.37 $\pm$ 0.35	39.12 $\pm$ 2.79	32.25 $\pm$ 0.76	30.61 $\pm$ 2.20
2 shots	36.75 $\pm$ 0.93	39.32 $\pm$ 1.54	38.93 $\pm$ 2.03	35.74 $\pm$ 2.39
4 shots	45.10 $\pm$ 1.11	45.96 $\pm$ 1.25	43.75 $\pm$ 1.35	43.59 $\pm$ 1.83
8 shots	48.99 $\pm$ 0.55	48.99 $\pm$ 0.84	47.44 $\pm$ 1.63	49.77 $\pm$ 1.75
16 shots	55.17 $\pm$ 0.68	57.65 $\pm$ 1.34	52.25 $\pm$ 0.78	55.44 $\pm$ 0.94
32 shots	62.24 $\pm$ 0.12	61.85 $\pm$ 0.76	59.32 $\pm$ 1.43	61.93 $\pm$ 1.21

TABLE 4. **Walnut Quality Grading — Fixed Data Augmentation** ($K = 6$ classes, Seeds 21, 22, 23). Zero-Shot CLIP Baseline: **22.26%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	35.28 $\pm$ 0.41	38.42 $\pm$ 0.75	33.02 $\pm$ 2.98	34.62 $\pm$ 1.11
2 shots	36.64 $\pm$ 1.20	41.76 $\pm$ 0.70	35.70 $\pm$ 1.71	36.21 $\pm$ 1.38
4 shots	43.05 $\pm$ 0.49	44.25 $\pm$ 1.69	41.49 $\pm$ 0.82	42.39 $\pm$ 1.06
8 shots	50.08 $\pm$ 0.68	48.95 $\pm$ 0.62	47.90 $\pm$ 1.40	49.73 $\pm$ 0.79
16 shots	55.21 $\pm$ 0.47	57.54 $\pm$ 0.41	55.32 $\pm$ 0.94	54.78 $\pm$ 1.12
32 shots	60.37 $\pm$ 0.65	62.20 $\pm$ 1.50	60.49 $\pm$ 1.23	60.14 $\pm$ 1.34

Note: The data augmentation pipeline was changed (fixed) in the 21, 22, 23 seed runs (letterbox square padding to eliminate object cropping, random vertical flips, translation shifts, and impulse/Gaussian noise).

TABLE 5. **Piarom Date Shape Grading — Initial Augmentation** ($K = 5$ classes, Seeds 11, 12, 13). Zero-Shot CLIP Baseline: **24.91%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	32.59 $\pm$ 2.05	28.00 $\pm$ 0.47	31.97 $\pm$ 5.72	33.40 $\pm$ 2.48
2 shots	34.82 $\pm$ 2.43	36.25 $\pm$ 1.03	39.71 $\pm$ 1.49	35.32 $\pm$ 1.96
4 shots	40.77 $\pm$ 3.02	44.55 $\pm$ 2.05	44.61 $\pm$ 1.83	42.13 $\pm$ 3.86
8 shots	52.23 $\pm$ 2.64	52.91 $\pm$ 1.99	48.27 $\pm$ 3.02	52.85 $\pm$ 1.72
16 shots	60.72 $\pm$ 0.60	62.08 $\pm$ 1.70	61.59 $\pm$ 0.84	62.45 $\pm$ 2.74
32 shots	67.91 $\pm$ 2.14	68.46 $\pm$ 0.39	67.91 $\pm$ 2.18	67.91 $\pm$ 1.14

TABLE 6. **Piarom Date Shape Grading — Fixed Data Augmentation** ($K = 5$ classes, Seeds 21, 22, 23). Zero-Shot CLIP Baseline: **24.91%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	30.42 $\pm$ 2.54	29.31 $\pm$ 0.47	29.43 $\pm$ 4.30	34.88 $\pm$ 2.00
2 shots	41.45 $\pm$ 0.85	40.77 $\pm$ 2.45	39.90 $\pm$ 3.02	41.88 $\pm$ 0.60
4 shots	48.08 $\pm$ 0.75	51.49 $\pm$ 1.16	48.27 $\pm$ 0.65	50.43 $\pm$ 2.14
8 shots	58.30 $\pm$ 2.09	60.22 $\pm$ 0.67	55.51 $\pm$ 1.06	57.37 $\pm$ 1.37
16 shots	67.60 $\pm$ 0.84	68.15 $\pm$ 1.41	64.87 $\pm$ 3.27	66.67 $\pm$ 1.14
32 shots	69.83 $\pm$ 0.60	70.32 $\pm$ 0.75	69.52 $\pm$ 1.70	69.27 $\pm$ 0.39

Note: The data augmentation pipeline was changed (fixed) in the 21, 22, 23 seed runs (letterbox square padding to eliminate object cropping, random vertical flips, translation shifts, and impulse/Gaussian noise).

TABLE 7. **Stanford Cars Classification** ($K = 196$ classes). Zero-Shot CLIP Baseline: **65.54%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	45.85 $\pm$ 0.63	69.30 $\pm$ 0.65	59.31 $\pm$ 0.93	45.74 $\pm$ 1.46
2 shots	56.93 $\pm$ 0.36	72.39 $\pm$ 0.28	64.85 $\pm$ 1.03	58.91 $\pm$ 0.83
4 shots	68.95 $\pm$ 0.67	76.34 $\pm$ 0.55	72.35 $\pm$ 0.81	70.05 $\pm$ 1.34
8 shots	76.85 $\pm$ 0.51	80.14 $\pm$ 0.56	78.82 $\pm$ 1.03	78.13 $\pm$ 1.11
16 shots	84.35 $\pm$ 0.28	84.89 $\pm$ 0.39	85.50 $\pm$ 0.15	85.33 $\pm$ 0.23
32 shots	87.68 $\pm$ 0.30	88.01 $\pm$ 0.22	88.50 $\pm$ 0.21	88.82 $\pm$ 0.06

2.3. Seed Sensitivity and Variance

Figure 2 shows the standard deviation of test accuracy across random seeds (11, 12, 13).

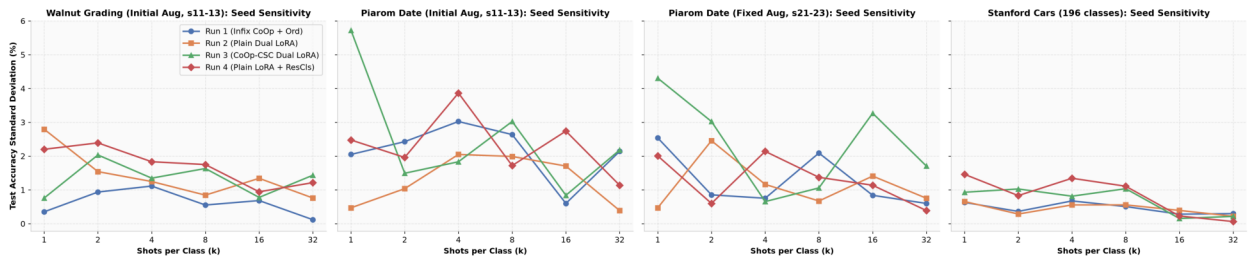


FIGURE 2. Test accuracy standard deviation across random seeds (11, 12, 13).

3. Training Accuracies

Figure 3 and Table 8 report the final training accuracy trajectories and exact numerical values (mean $\pm$ standard deviation across seeds) across shot counts ($k \in \{1, 2, 4, 8, 16, 32\}$) for each evaluated paradigm. The top-performing method at each shot is highlighted in green.

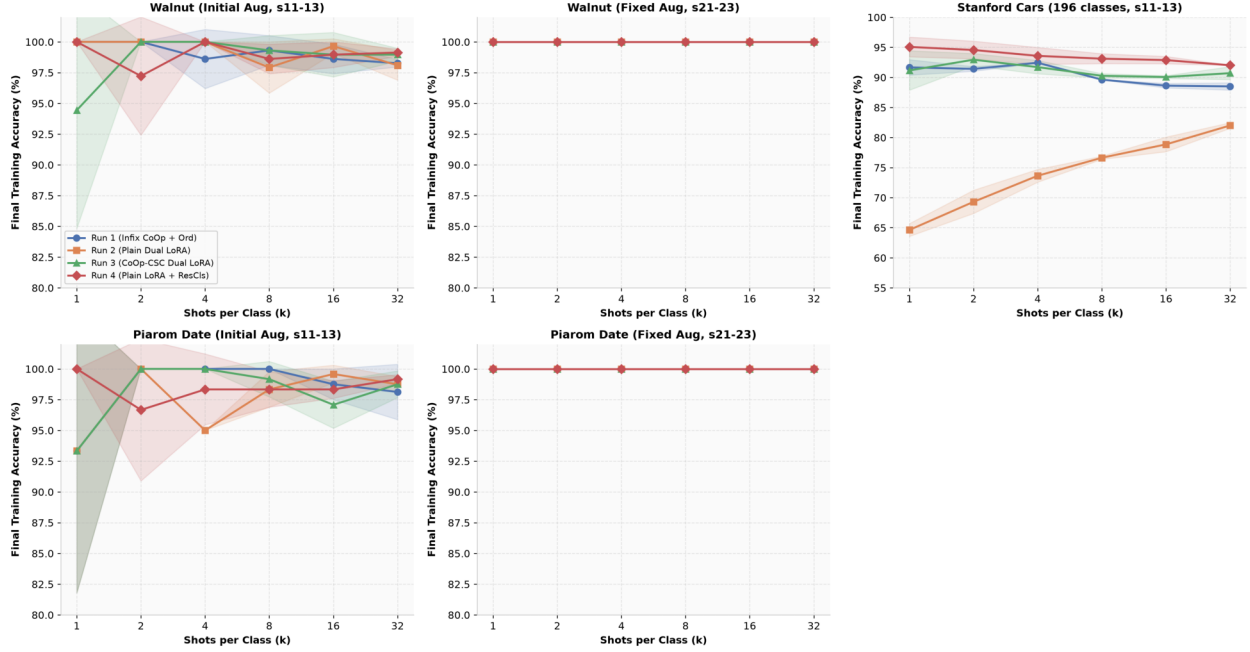


FIGURE 3. Few-shot final training accuracy trajectories across 1 to 32 shots on Walnut under initial and fixed data augmentations ($K = 6$), Pirom Date Shape under initial and fixed data augmentations ($K = 5$), and Stanford Cars ($K = 196$). Shaded bands represent ± 1 standard deviation.

TABLE 8. **Final Training Accuracy Across Benchmarks** (mean $\pm$ standard deviation across seeds).

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
Walnut Quality Grading — Initial Augmentation ($K = 6$ classes, Seeds 11, 12, 13)				
1 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	94.44 $\pm$ 9.62	100.00 $\pm$ 0.00
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	97.22 $\pm$ 4.81
4 shots	98.61 $\pm$ 2.41	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
8 shots	99.31 $\pm$ 1.20	97.92 $\pm$ 2.08	99.31 $\pm$ 1.20	98.61 $\pm$ 1.20
16 shots	98.61 $\pm$ 1.20	99.65 $\pm$ 0.60	98.96 $\pm$ 1.80	98.96 $\pm$ 1.04
32 shots	98.26 $\pm$ 0.30	98.09 $\pm$ 1.20	98.96 $\pm$ 0.52	99.13 $\pm$ 0.30
Walnut Quality Grading — Fixed Augmentation ($K = 6$ classes, Seeds 21, 22, 23)				
1 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
4 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
8 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
16 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
32 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
Piarom Date Shape Grading — Initial Augmentation ($K = 5$ classes, Seeds 11, 12, 13)				
1 shots	93.33 $\pm$ 11.55	93.33 $\pm$ 11.55	93.33 $\pm$ 11.55	100.00 $\pm$ 0.00
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	96.67 $\pm$ 5.77
4 shots	100.00 $\pm$ 0.00	95.00 $\pm$ 0.00	100.00 $\pm$ 0.00	98.33 $\pm$ 2.89
8 shots	100.00 $\pm$ 0.00	98.33 $\pm$ 1.44	99.17 $\pm$ 1.44	98.33 $\pm$ 1.44
16 shots	98.75 $\pm$ 1.25	99.58 $\pm$ 0.72	97.08 $\pm$ 1.91	98.33 $\pm$ 0.72
32 shots	98.12 $\pm$ 2.25	98.75 $\pm$ 0.62	98.75 $\pm$ 1.08	99.17 $\pm$ 0.36
Piarom Date Shape Grading — Fixed Augmentation ($K = 5$ classes, Seeds 21, 22, 23)				
1 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
4 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
8 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
16 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
32 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
Stanford Cars Classification ($K = 196$ classes, Seeds 11, 12, 13)				
1 shots	91.67 $\pm$ 1.28	64.63 $\pm$ 1.06	91.16 $\pm$ 3.28	95.07 $\pm$ 1.64
2 shots	91.41 $\pm$ 0.39	69.30 $\pm$ 1.93	92.94 $\pm$ 1.03	94.56 $\pm$ 1.47
4 shots	92.43 $\pm$ 0.07	73.64 $\pm$ 1.06	91.71 $\pm$ 1.11	93.58 $\pm$ 1.40
8 shots	89.63 $\pm$ 0.04	76.64 $\pm$ 0.26	90.26 $\pm$ 0.36	93.11 $\pm$ 0.83
16 shots	88.63 $\pm$ 0.37	78.86 $\pm$ 1.22	90.06 $\pm$ 0.23	92.88 $\pm$ 0.61
32 shots	88.50 $\pm$ 0.61	82.00 $\pm$ 0.46	90.70 $\pm$ 1.08	92.02 $\pm$ 0.02